

## **Release of erythropoietin and neuron-specific enolase after breath-hold in competing free divers**

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**Running head:** Biomarkers of hypoxia after apnoea

### **Key words:**

Apnoea, brain, erythropoietin, hypoxia, heart, kidney, NSE

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## Abstract

*Background:* Free diving is associated with extreme hypoxia. This study evaluated the combined effect of maximal static breath hold and underwater swimming on plasma biomarkers of tissue hypoxemia: erythropoietin, neuron-specific enolase and S100B, C-reactive protein, pro-atrial natriuretic peptide and troponin T.

*Methods:* Venous blood samples were obtained from seventeen competing free divers before and 3 hours after sessions of static apnoea and underwater swimming. The heart was evaluated by echocardiography.

*Results:* Static apnoea for  $293 \pm 78$  s (mean  $\pm$  SD) and subsequent  $88 \pm 21$  m underwater swimming increased plasma erythropoietin from  $10.6 \pm 3.4$  to  $12.4 \pm 4.1$  mIU/l ( $P = 0.013$ ) and neuron-specific enolase from  $14.5 \pm 5.3$  to  $24.6 \pm 6.4$  ng/ml ( $P = 0.017$ ); C-reactive protein decreased from  $0.84 \pm 1.0$  to  $0.71 \pm 0.67$  mmol/l ( $P = 0.013$ ). In contrast, plasma concentrations of S100B ( $P = 0.394$ ), pro-atrial natriuretic peptide ( $P = 0.549$ ) and troponin T ( $P = 0.125$ ) remained unchanged and, as assessed by echocardiography, the heart was not affected.

*Conclusion:* In competitive free divers, bouts of static and dynamic apnoea increase plasma erythropoietin and neuron specific enolase, suggesting that renal and neural tissue, rather than the heart is affected by the hypoxia developed during apnoea and underwater swimming.

## Introduction

Competitive free diving has evolved since Charles de Vendeville won the 1900 Olympic Games accomplishing 60 m in a pool. Free-divers now hyperinflate their lungs (Eichinger et al. 2008), which enables them to hold their breath for more than 10 min and swim as far as 250 m or go below 200 m in depth (AIDA international), but they are at risk of ‘shallow water blackout’ or loss of consciousness and also loss of motor control, pulmonary barotrauma, and even death (AIDA international; Schiffer and Lindholm 2013). Complications associated with free diving are probably related to bradycardia, reduced cardiac output and hypoxemia (Batinic et al. 2011; Lindholm 2007). Thus, during a breath hold, muscle oxygenation becomes low (Kjeld et al. 2014) whereas cerebral perfusion augments as hypercapnia develops (Kjeld et al. 2009). Competitive divers are characterized by attenuated diving response, post-apnoea acidosis, and oxidative stress (Bulmer et al. 2008) and also reduced sensitivity to hypoxia and hypercapnia (Foster and Sheel 2005). Yet, it may be that free divers after apnoea elicits both heart and brain damage as a consequence of pronounced hypoxemia (Andersson et al. 2009a; Andersson et al. 2009b).

We hypothesised that biomarkers of hypoxemia would indicate which organs, if any, are affected by breath hold and underwater swimming. In competitive free divers plasma troponin T and pro-atrial natriuretic peptide (proANP) were determined to indicate whether the heart becomes affected, S100 calcium binding protein B (S100B) and neuron-specific enolase (NSE) to indicate an effect on neuronal tissue, erythropoietin to indicate overall and renal affection, and C-reactive protein to indicate inflammation. Also the divers underwent an echocardiographic evaluation following the underwater swim.

## Methods

While participating in the Danish free diving championship seventeen healthy non-smoking free divers (Table 1) provided oral and written consent for the study as approved by the Regional Ethical Committee of the Capital Region of Denmark (H-3-2012-058). Five of the participants were in the top 10 of the World Ranking List and one was an indoor free diving World champion.

### *Protocol*

For one day prior to the study, the volunteers refrained from planned physical exercise and consumption of alcohol. The volunteers prepared for the competition according to the AIDA regulations (AIDA international): After hyperventilation and glossopharyngeal insufflation, the volunteers performed one maximal static breath hold in a 28° C 0.8 m deep indoor pool (n = 15). After  $306 \pm 50$  min (range 206-414 min), the volunteers repeated hyperventilation and glossopharyngeal insufflation and swam underwater for as long as possible in a 26° C 50 m, 2 m deep indoor pool (2 divers decided to perform only the underwater swimming). During both breath holds the volunteers were wearing a diver's mask, fins, a neoprene suit, and a 3.5 kg balancing lead. No volunteer used ischemic preconditioning (Kjeld et al. 2014) before the breath hold or underwater swimming. The efforts were video-recorded and a safety swimmer and a physician observed the sessions.

### *Biochemical analyses*

Blood samples (5 ml) were obtained from an arm vein before and  $182 \pm 52$  min (range 94-257 min) after the first apnoea. Blood was placed in ice water and centrifuged (800–1000 *g*) for 10 min and plasma stored at -80° C until analysis.

Cardiac high sensitive troponin T (Thygesen et al. 2010) and proANP (De Palo et al. 2000; Goetze et al. 2012) were determined to indicate affection of the heart, while C-reactive protein indicated inflammation and risk for coronary heart disease (Buckley et al. 2009; Ridker 2003). To evaluate whether neuronal tissue became affected S100B (Nygaard et al. 1997) and NSE (Schmechel et al. 1978) were determined. Cardiac troponin T was measured by a Roche Elecsys 2010 analyzer (Elecsys Troponine T STAT Immunoassay, third generation; Roche Diagnostics, Mannheim, Germany) (Klein et al. 1998) with a detection limit below 0.01 µg/L. C-reactive protein was determined by the Roche/Hitachi Tina-quant (Latex) assay (Hitachi 917; Roche Diagnostics) (Eda et al. 1998). Imprecision was < 6% at <1.0 mg/L. Analysis of N-terminal proANP was performed by an in-house immunoassay. The assay procedure included plasma extraction followed by tryptic cleavage, which releases the assay epitope from the N-terminal region (Goetze et al. 2012; Hunter et al. 2011); the limit of detection was 34 pmol/L. S100B was determined using the Roche Elecsys<sup>®</sup> S100 reagent kit (Roche Diagnostics) (Nygaard et al. 1997), measuring range 0.005–39 µg/L. NSE was measured by an electrochemiluminescent assay using reagent sets (Roche Diagnostics) and the Elecsys 2010 analyser; the limit of detection was 0.05 ng/mL. Erythropoietin was determined by a two-site sandwich immunoassay with chemiluminescent detection on an automated random access immunoassay analyser (Immulite; DPC, R & D systems Inc., Minneapolis) (Mossuz et al. 2005) with reference interval equal to 4.1–20.1 IU/L.

### *Transthoracic echocardiography*

After the static and dynamic apnoea, transthoracic echocardiography (Evangelista et al. 2008) was performed with a Vivid 7 machine (GE Healthcare, Wisconsin) and results evaluated by a cardiologist. Both investigators were blinded to the biochemical results. The evaluation included ejection fraction and stroke volume using the end systolic and diastolic volumes, cardiac output multiplying stroke volume and heart rate, left atrial volume index using a biplane area-length

formula, end-diastolic inter-ventricular septal, posterior wall thickness, left ventricle out tract diameter, aortic sinus diameter, left ventricular internal dimension in diastole, right ventricular function using tricuspid annular plane excursion, aortic valve maximal velocity and peak gradient, and tricuspid valve maximal peak gradient. Diastolic variables included peak early and late diastolic mitral inflow velocity and their ratio, deceleration time, and average of the medial and lateral mitral annular diastolic velocities were determined, based on the mean of two consecutive cardiac cycles.

### *Statistical analysis*

Variables are presented as means  $\pm$  SD. Wilcoxon's signed test by rank was used to analyse changes in plasma variables before and after the apnoea. Pearson's correlation coefficient was used to test the association between changes in biomarkers and duration of apnoea. A P-value  $< 0.05$  was considered statistically significant.

### Results

**The static apnoea lasted  $293 \pm 78$  s and the divers covered  $88 \pm 21$  m during underwater swimming lasting  $118 \pm 36$  s. The total apnoea time was 537 (range 100 – 637) s. During the static apnoea nine divers developed loss of motor control and one loss of consciousness, while four participants demonstrated loss of motor control and another four developed loss of consciousness during the dynamic apnoea. Plasma concentrations of proANP (P = 0.549), cardiac troponin T (P = 0.125) and S100B (P = 0.394) remained unchanged in response to the apnoea (Table 2). Only for one subject could troponin T be detected before the breath holds (3.57 pg/ml) that increased to 8.94 pg/ml after the dive and troponin T was detected for further for two volunteers (5.72 and 10.85 pg/ml, respectively) after the dive. However, combined static breath hold and underwater swimming increased plasma erythropoietin (P = 0.013) and neuron-specific enolase (Table 2, Figure 1). At the same**

time, there was a decrease in plasma C-reactive protein ( $P = 0.013$ , Table 2). A subgroup analysis revealed no difference in plasma markers when comparing those participants who developed loss of consciousness or of motor control ( $n = 9$ ) with those who did not ( $n = 8$ , data not shown). Furthermore, there was no association between the duration of static or dynamic apnoea or their sum and the changes in S100B ( $r = 0.269$ ;  $P = 0.296$ ), NSE ( $r = 0.208$ ;  $P = 0.424$ ), troponin T ( $r = -0.0799$ ;  $P = 0.777$ ), proANP ( $r = -0.167$ ;  $P = 0.553$ ) or C-reactive protein ( $r = 0.140$ ;  $P = 0.618$ ). However, the increase in erythropoietin demonstrated a weak negative correlation to the apnoea time ( $r = -0.52$ ;  $P = 0.045$ ). Transthoracic echocardiographic findings were all normal (Table 3): All volunteers demonstrated normal global left ventricular function and no heart valve disease was detected according to the recommendations of European Association of Echocardiography for standardization of performance, digital storage and reporting of echocardiographic studies (Evangelista et al. 2008).

## Discussion

This study used plasma biomarkers to indicate which organs, if any, become affected when elite free divers perform static apnoea and underwater swimming. Increased plasma erythropoietin was indicative of renal hypoxia and also the nervous system became affected as indicated by increased plasma levels of the brain injury marker NSE. In contrast, levels of plasma proANP and troponin T did not change significantly, and also echocardiography after the apnoea was normal, indicating no affection of the heart. Furthermore, the decrease in C-reactive protein suggests that inflammation was not involved.

Nine divers developed loss of consciousness or loss of motor control, but no correlation to the cardiac biomarkers were observed in these subjects. Hence, the findings do not support that hyperinflation of the lungs and apnoea affects the heart (Andersson et al. 2009a). It was not possible to evaluate oxygen saturation during the apnoea, but a  $\sim 3$  min static apnoea reduces arterial oxygen

saturation by ~ 33% (Kjeld et al. 2009), muscle oxygenation by ~ 57%, and cerebral oxygenation by ~ 44% (Kjeld et al. 2014). In the present study volunteers performed apnea of longer duration and hence the saturation would be assumed to be well below these values.

Both static and dynamic apnoea induce rapid and profound cardiovascular changes including bradycardia, reduction of cardiac output and redistribution of regional blood flow (Alboni et al. 2011; Breskovic et al. 2011; Richardson et al. 2012). Blood flow to the kidneys and gut virtually ceases (Bron et al. 1966) while cerebral perfusion increases (Kjeld et al. 2009). Thus, the diving response serves as an oxygen-conserving redistribution of blood to the brain at the expense of perfusion to the renal and splanchnic vascular beds (de Bruijn et al. 2009).

Combined bouts of maximum static breath hold and underwater swimming provoked a modest, albeit significant increase in plasma erythropoietin as also found by de Bruijn et al. (2009) in volunteers undergoing series of maximal apnoea. The current increase in plasma erythropoietin was much lower than the peak value seen 18 hours after arrival at 4350 m altitude (Klausen et al. 1996) but equals that developed in elite runners living at 3000 m altitude and training at sea level (Brugniaux et al. 2006).

As reported after scuba diving (Stavrinou et al. 2011) but in contrast to earlier studies (Andersson et al. 2009b; Liner and Andersson 2009), we found that apnoea and underwater swimming did not change the circulating level of plasma S100B. In the study by Andersson et al. (2009b), blood samples were collected 15, 30, 60, and 120 min after apnoea of up to 403 s and S100B peaked after 30 min reaching up to 167 % of baseline level and decreased to baseline after 120 min. In our study blood samples were collected, on average, three hours after the apnoea and increases in S100B, if any, are therefore likely to have been missed. This study showed that competitive breath hold diving increases plasma NSE to levels found in survivors from cardiac arrest with a good neurological outcome (Schoerhuber et al. 1999). NSE is the neuronal form of



the glycolytic enzyme enolase (2-phospho-D-glycerate hydrolase) that is located almost exclusively within neurons and neuroendocrine cells (Schmechel et al. 1978) and plasma NSE reflects parenchymal tissue injury (Schoerhuber et al. 1999). Thus, competitive free divers might expose themselves to potential brain injury. Yet, even years of exposure to breath hold diving seems without effect on neuropsychological tests (Ridgway and McFarland, 2006), but as these tests were carried out before today's extreme durations of apnea, there is a need for re-evaluation.

There are several limitations to this observational study of breath-hold divers during a competition that excluded controlled and intervening experiments. A distinction between static and dynamic effects of apnoea was not elaborated. Measurements concerned plasma biomarkers and their relevance to long-term outcome needs to be established (Ridgway and McFarland, 2006). The markers of neuronal damage were measured peripherally, not across the brain, and particularly in the case of NSE the origin cannot be known for certain. The present ELISA for analysis of EPO is not specific for kidney EPO, so the current results reflect changes from all potential sources of endogenous EPO. However, in adults more than 90% of circulating EPO originates from the kidneys. In addition, differences related to age or gender cannot be ruled out.

In conclusion, 9 of 17 competitive free divers developed black out after apnoea and underwater swimming and increased plasma values of neuron-specific enolase implies that the brain was affected. Also, the kidneys may be affected as indicated by the increase in plasma erythropoietin. On the other hand, unchanged proANP and troponin T as well as a decrease in C-reactive protein and a normal echocardiographic evaluation indicate no affection of the heart.

## **Perspectives**

After apnoea and underwater swimming plasma erythropoietin was comparable to the level demonstrated in elite runners living at an altitude of 3000 m and training at sea level (Brugniaux et

al. 2006). Thus, repeated apnoea would be expected to enhance red blood cell mass and be of benefit for endurance athletes, albeit it needs to be evaluated to what extent apnoea enhances the hematocrit. Furthermore, the results suggest that apnoea does not affect the heart supporting that diving might be of interest to those seeking enhanced aerobic capacity. Yet, the plasma levels of NSE indicate that hypoxia affected the nervous system (Schoerhuber et al. 1999; Andersson et al. 2009b; Liner and Andersson 2009; Ridgway et al. 2006) which may be problematic in regard to advocate diving including glossopharyngeal insufflation as a mean to increase red blood cell mass. Repetitive apnoea may reduce the chemoreceptor responsiveness to CO<sub>2</sub> and this may limit the ventilatory response to exercise. Thus, specific studies will be needed in volunteers to determine if apnoea may be more deleterious than advantageous as concurrent “training” for endurance athletes.

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## Conflicts of interest

The authors declare no conflict of interest.

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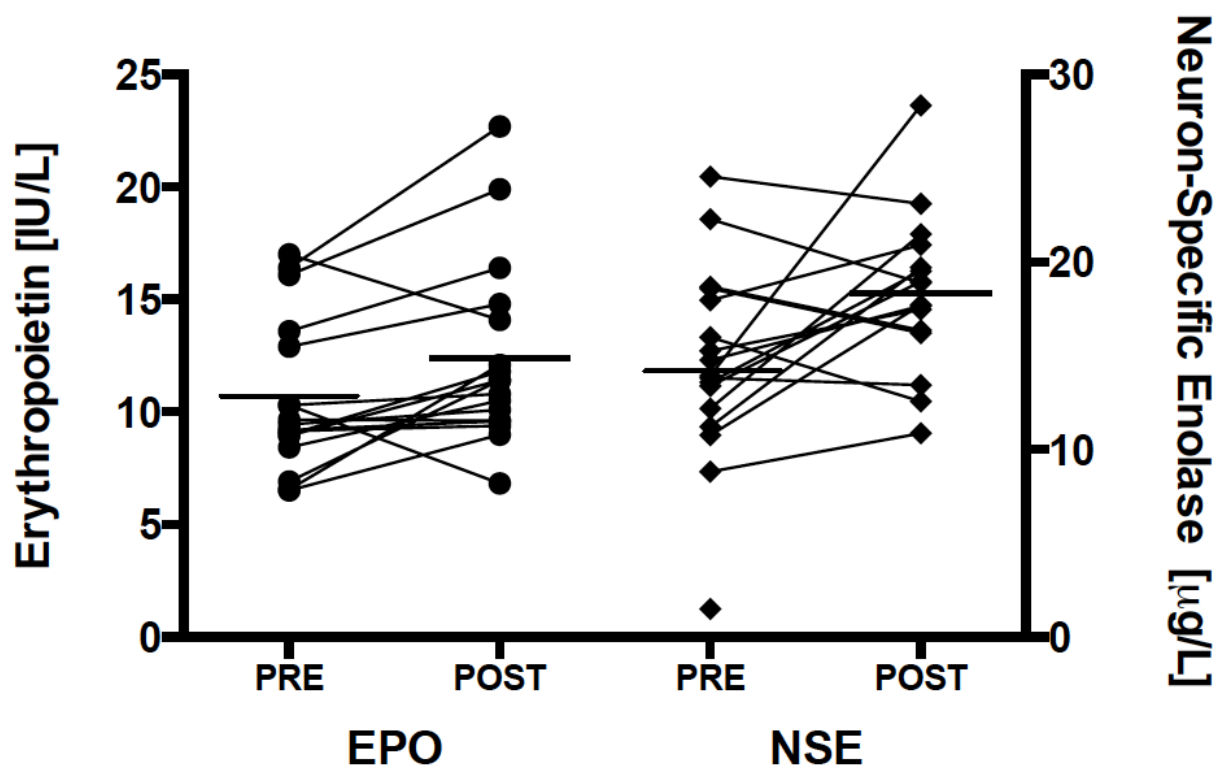
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**Table 1.** Subject characteristics.

|                                      | <b>Divers</b>  |
|--------------------------------------|----------------|
| No. subjects                         | 17 (6 females) |
| Age (years)                          | 32 ± 7         |
| Height (cm)                          | 178 ± 8        |
| Weight (kg)                          | 72 ± 10        |
| Systolic blood pressure (mm Hg)      | 121 ± 15       |
| Diastolic blood pressure (mm Hg)     | 70 ± 11        |
| Mean arterial blood pressure (mm Hg) | 87 ± 12        |
| Heart Rate (bpm)                     | 66 ± 8         |

Values are mean ± SD.

Figure 1. Erythropoietin (EPO) and neuron specific enolase (NSE) before and after maximal static ( $n=15$ ) and dynamic ( $n=17$ ) breath hold. Horizontal lines represent means. Both EPO and NSE on average increased significantly ( $P=0.013$  and  $P=0.017$ , respectively).





**Table 2.** Pro-atrial natriuretic peptide (proANP), high sensitive troponin T (hsTrop T), C-Reactive protein (CRP), neurospecific enolase (NSE), S100B, and erythropoietin (EPO) before and after maximal static ( $n=15$ ) and maximal dynamic ( $n=17$ ) breath hold.

|                  | Rest            | Post dive         |
|------------------|-----------------|-------------------|
| proANP, pmol/l   | $275 \pm 66$    | $287 \pm 89$      |
| hsTrop T, pg/ml  | $3.0 \pm 0.1$   | $4.1 \pm 2.4$     |
| CRP, mg/L        | $0.84 \pm 1.0$  | $0.71 \pm 0.67$ * |
| NSE, ng/ml       | $14.5 \pm 5.3$  | $24.6 \pm 6.4$ *  |
| S100B, $\mu$ g/l | $0.10 \pm 0.13$ | $0.07 \pm 0.03$   |
| EPO, mIU/l       | $10.6 \pm 3.4$  | $12.4 \pm 4.1$ *  |

Values are mean  $\pm$  SD. \*:  $P < 0.05$  compared to rest.

**Table 3.** Echocardiographic characteristics ( $n=15$ ).

|                  |                  |                                   |             |                  |             |
|------------------|------------------|-----------------------------------|-------------|------------------|-------------|
| <b>Dimension</b> | <b>LA</b>        | Volume/<br>BSA, ml/m <sup>2</sup> | 14.8 ± 3.9  |                  |             |
|                  | <b>LV</b>        | IVSDd, cm                         | 0.9 ± 0.2   | ESV, ml          | 33.4 ± 9.0  |
|                  |                  | LVIDd/BSA,<br>cm/m <sup>2</sup>   | 2.5 ± 0.3   | EDV, ml          | 87.7 ± 22.7 |
|                  |                  | LVPWd, mm                         | 0.9 ± 0.1   |                  |             |
|                  | <b>Aorta</b>     | LVOT, cm                          | 2.1 ± 0.2   | Sinus, cm        | 2.6 ± 0.3   |
| <b>Function</b>  | <b>LV</b>        | LVEF, %                           | 62 ± 3      | CO, l/min        | 3.8 ± 1.2   |
|                  |                  | SV, ml                            | 54.6 ± 14.3 |                  |             |
|                  | <b>RV</b>        | TAPSE, cm                         | 2.4 ± 0.2   |                  |             |
| <b>Valves</b>    | <b>Aorta</b>     | V2 max, m/s                       | 1.2 ± 0.3   | max PG,<br>mm Hg | 6.4 ± 4.1   |
|                  | <b>Mitral</b>    | E/A ratio                         | 1.2 ± 0.5   | dec slope, s     | 4.4 ± 1.2   |
|                  | <b>Tricuspid</b> | max PG,<br>mm Hg                  | 13.9 ± 8.9  |                  |             |

LA = left atrium; BSA= body surface area (Dubois); LV = left ventricle; IVSDd= inter ventricular septum diameter diastolic, LVIDd/BSA = left ventricle internal diameter diastolic (values were corrected for body surface area); LVPWd = left ventricle posterior wall diastolic diameter; ESV= end systolic volume; EDV= end diastolic volume; LVOT = left ventricle out tract; LVEF = left ventricle ejection fraction (Simpson); SV= stroke volume; CO= cardiac output; dec slope = mitral valve deceleration slope; RV= right ventricle; TAPSE = *Tricuspid annular plane systolic excursion*; V2 max = aortic valve max velocity; max PG = max peak gradient.

Values are means ± SD.